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(54) **FORMATION FLUID SAMPLING APPARATUS AND METHODS**

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E21B 49/00 (2006.01)

(52) **U.S. Cl.** **166/264**; 166/100; 175/59

(58) **Field of Classification Search** 166/100, 166/264; 175/50, 59; 73/152.19, 152.24, 73/152.26

See application file for complete search history.

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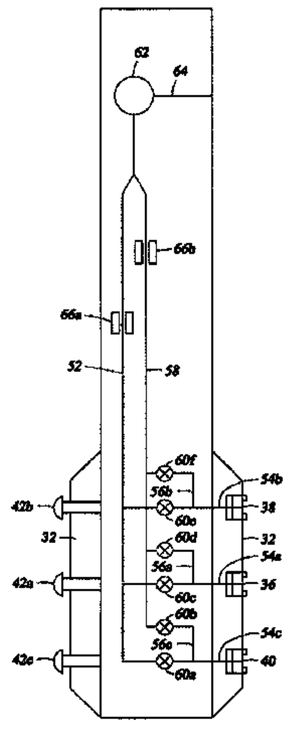
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(57) **ABSTRACT**

A fluid sampling system retrieves a formation fluid sample from a formation surrounding a wellbore extending along a wellbore axis, wherein the formation has a virgin fluid and a contaminated fluid therein. The system includes a sample inlet, a first guard inlet positioned adjacent to the sample inlet and spaced from the sample inlet in a first direction along the wellbore axis, and a second guard inlet positioned adjacent to the sample inlet and spaced from the sample inlet in a second, opposite direction along the wellbore axis. At least one cleanup flowline is fluidly connected to the first and second guard inlets for passing contaminated fluid, and an evaluation flowline is fluidly connected to the sample inlet for collecting virgin fluid.

26 Claims, 8 Drawing Sheets



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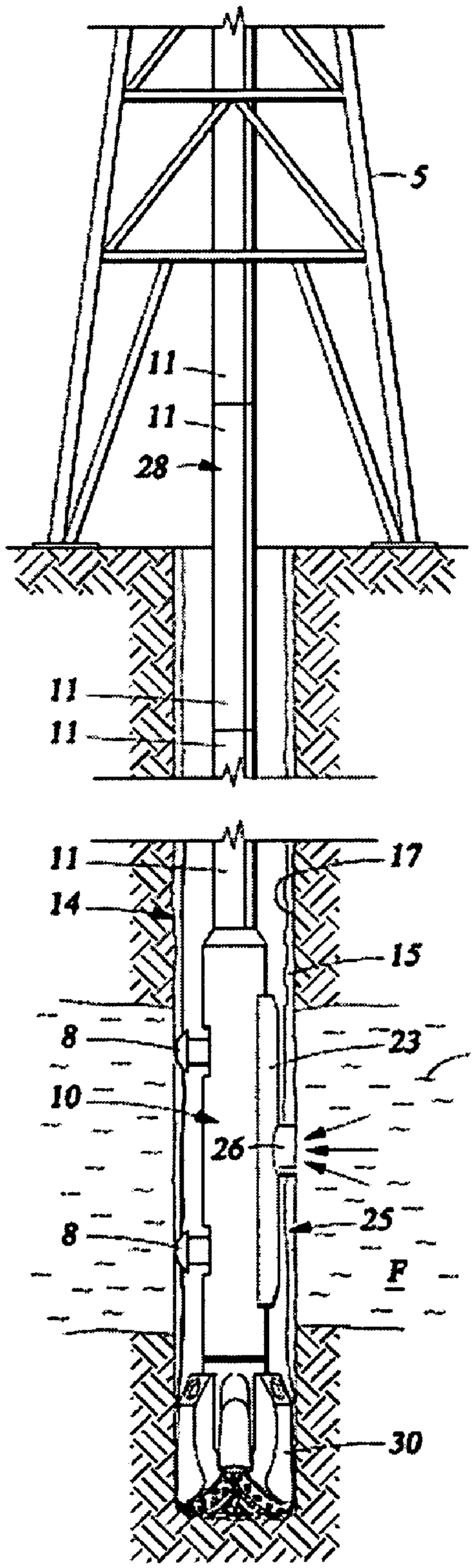

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Fig. 1

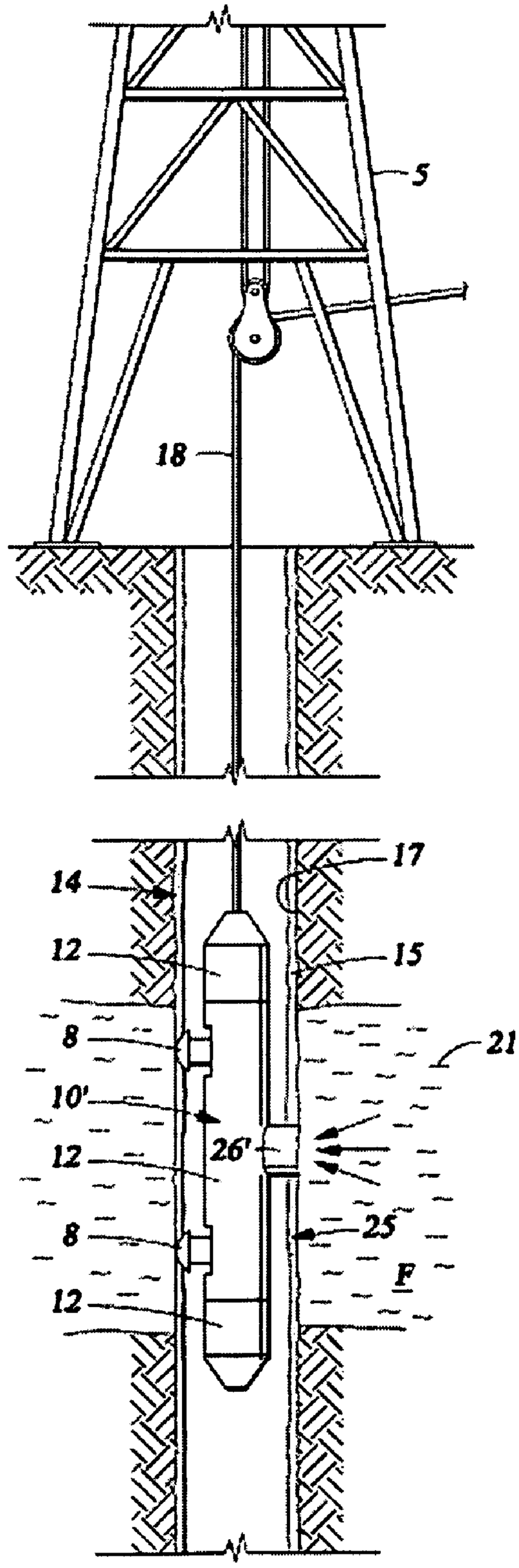

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Fig. 2

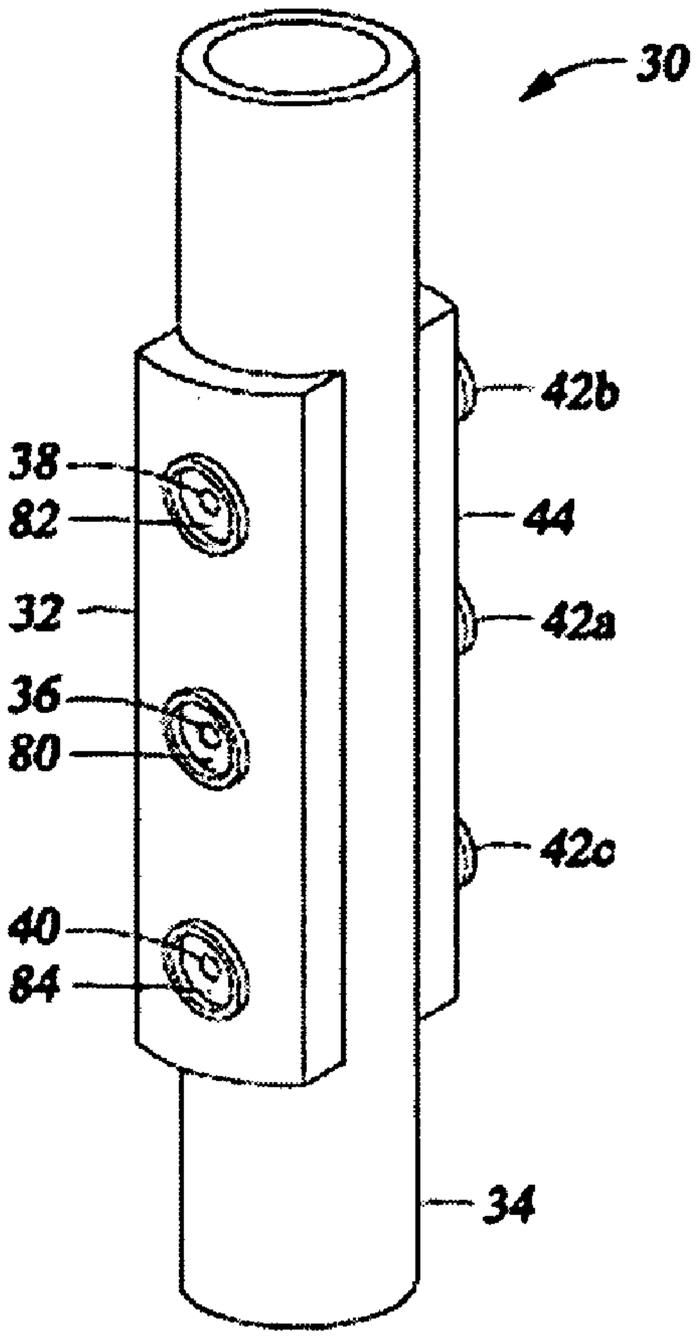

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Fig. 3

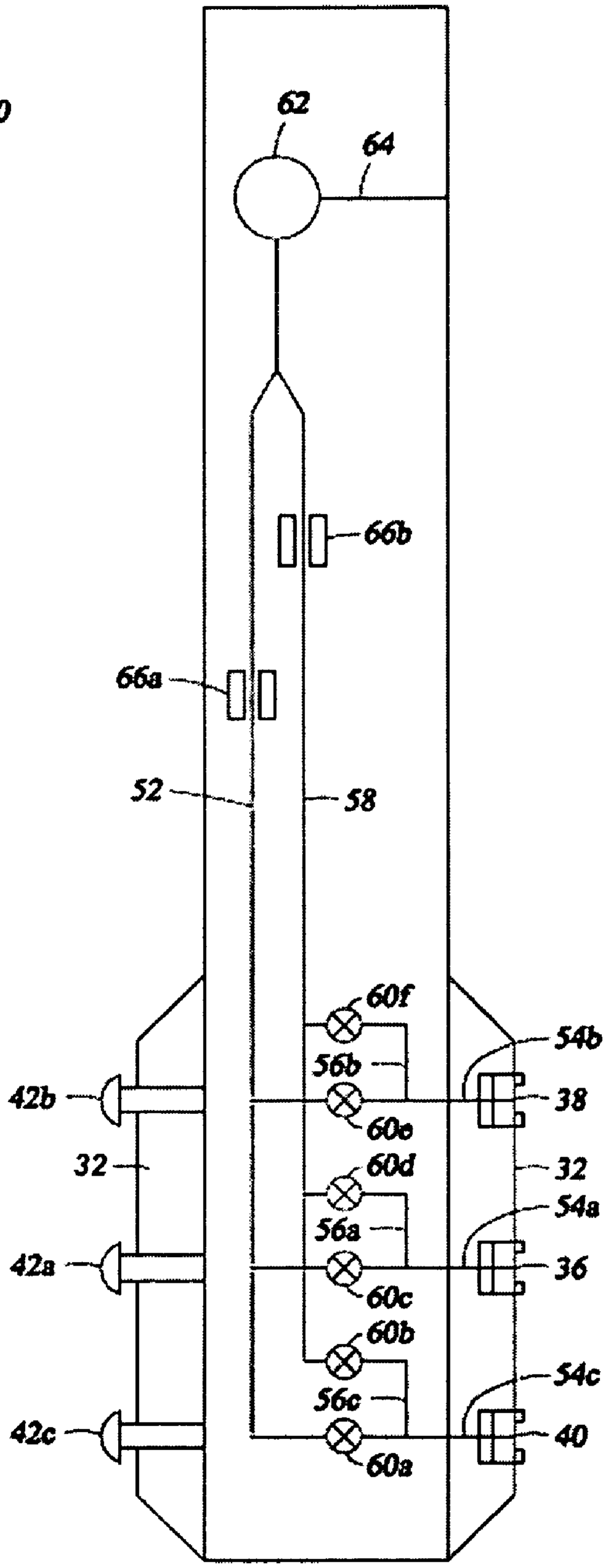

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Fig. 4

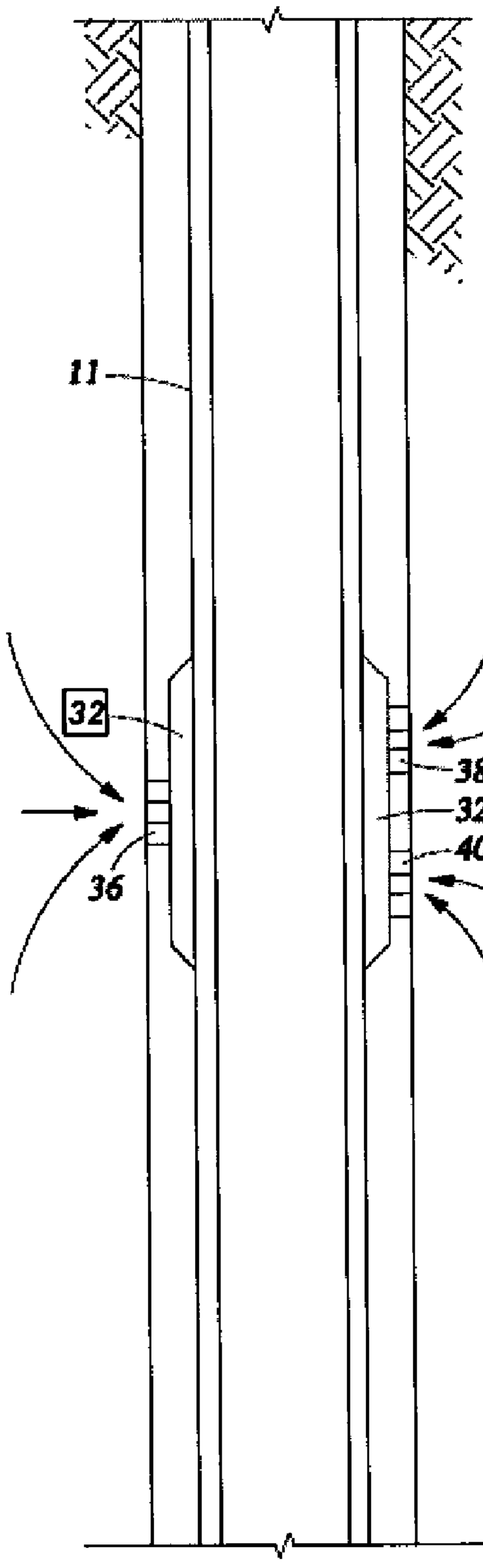


Fig. 5

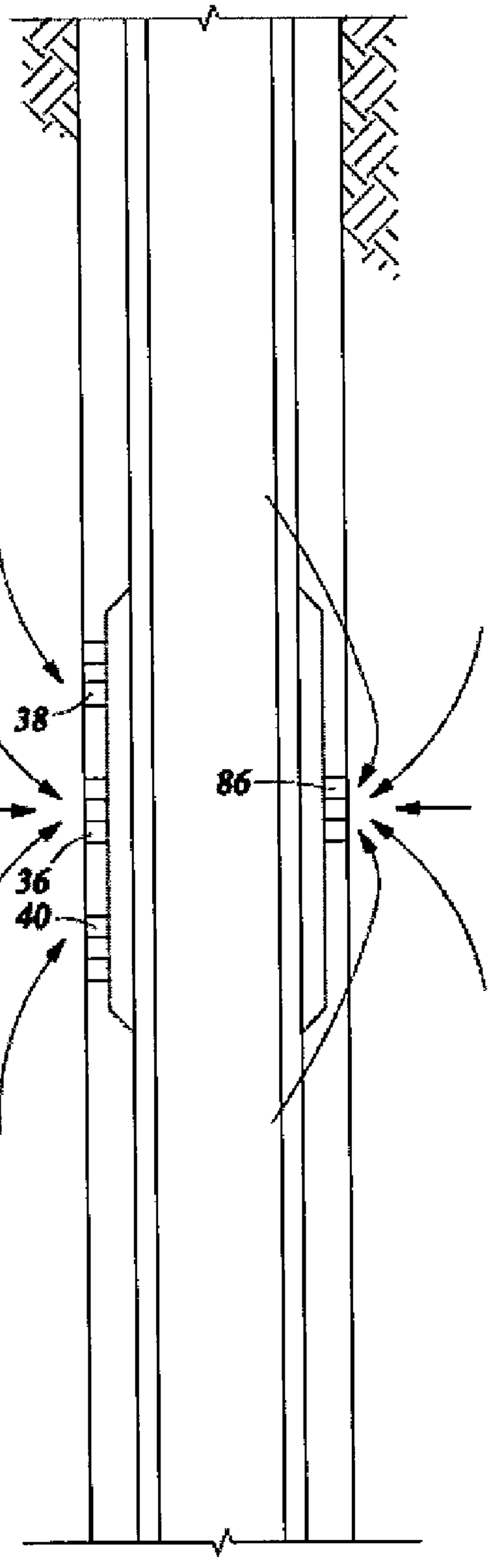


Fig. 6

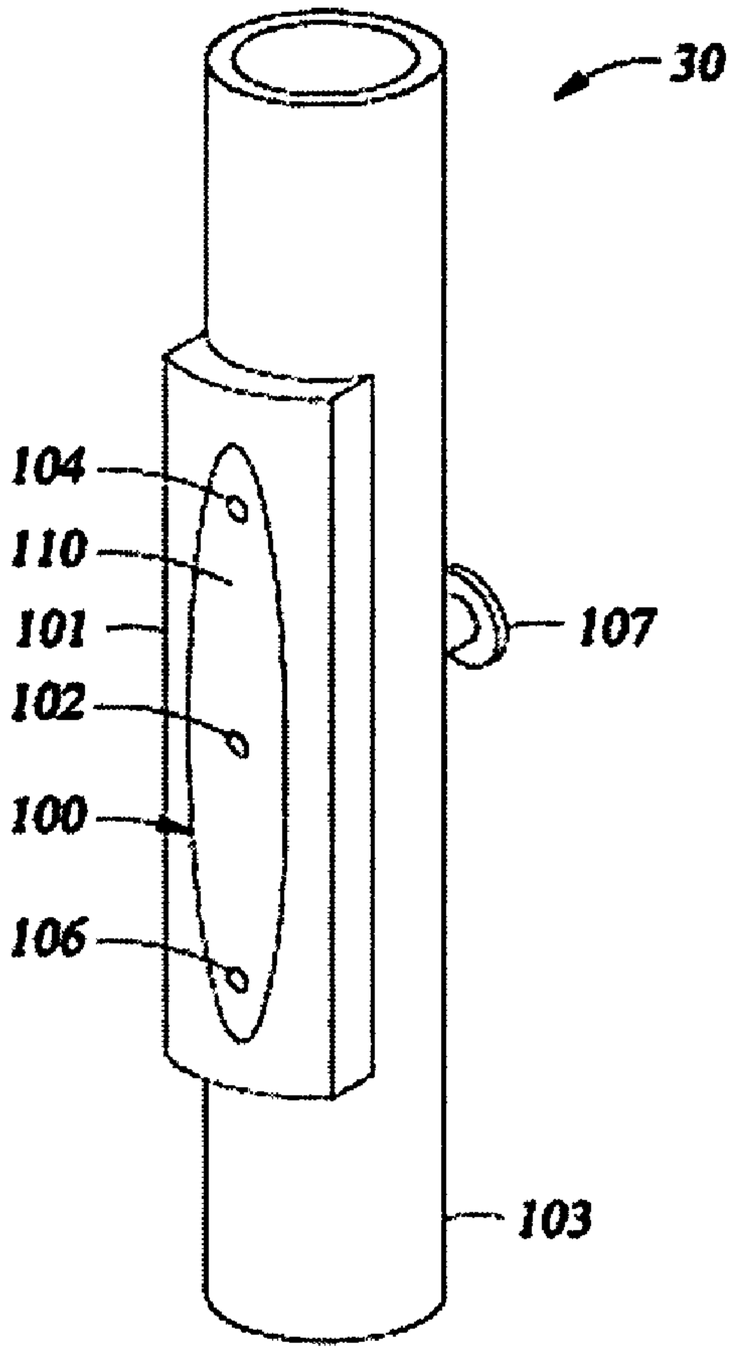


Fig. 7

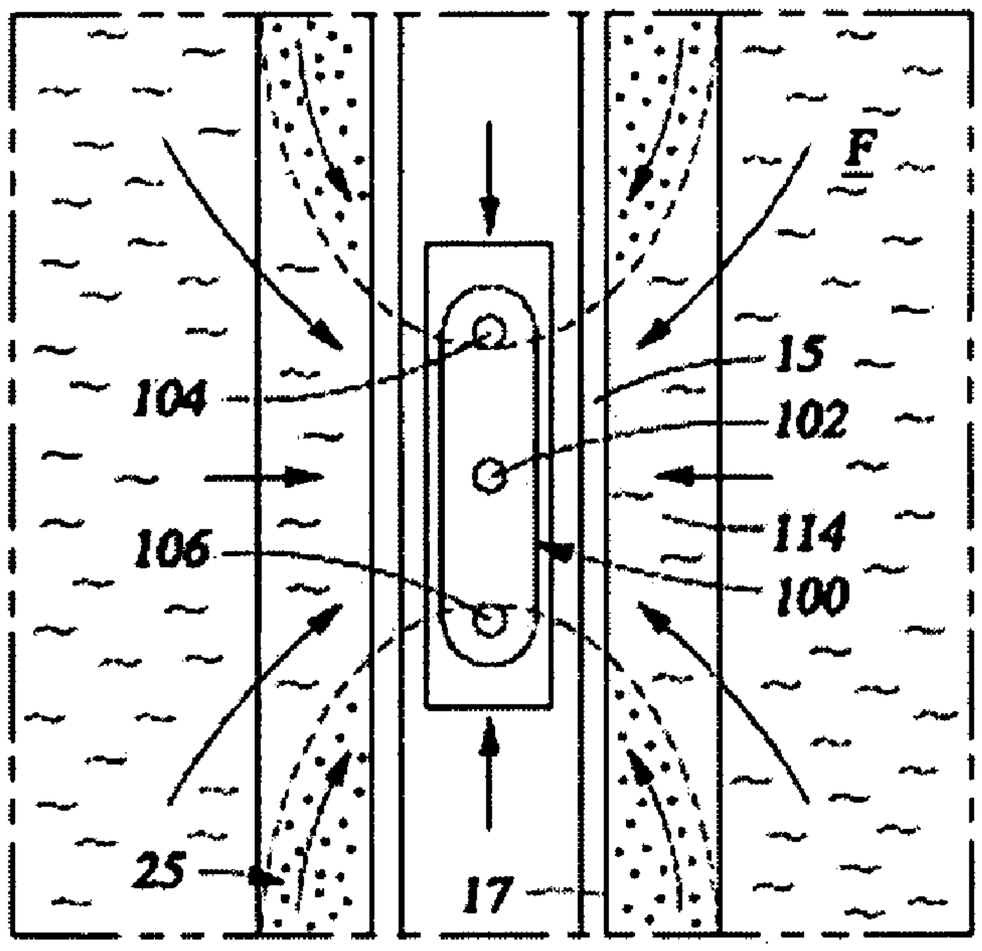


Fig. 8

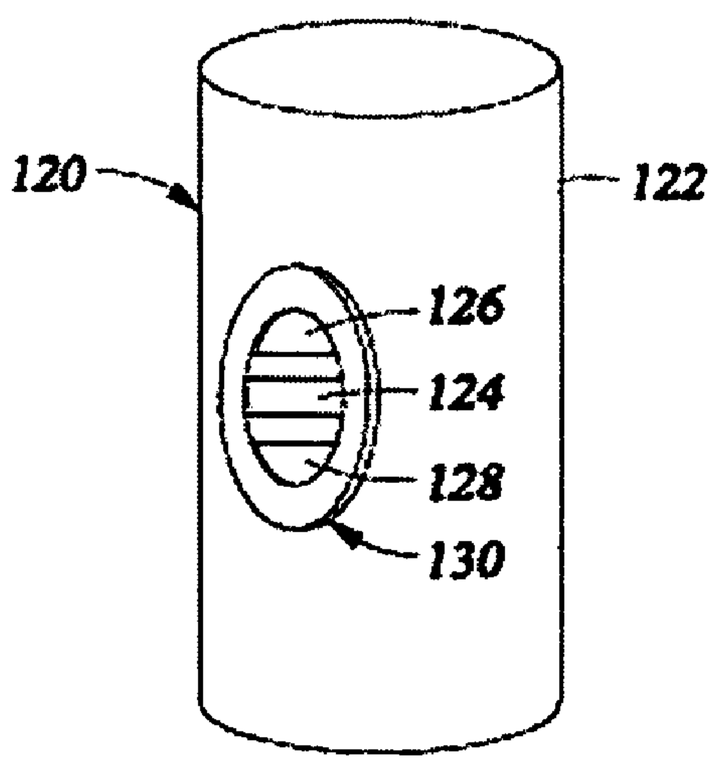


Fig. 9

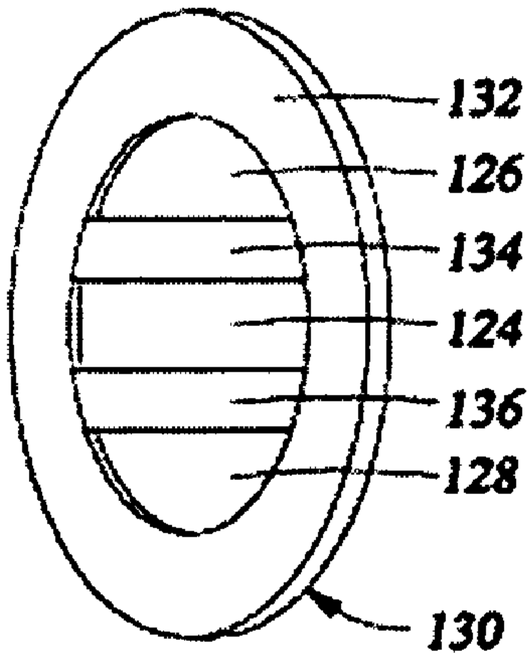


Fig. 10

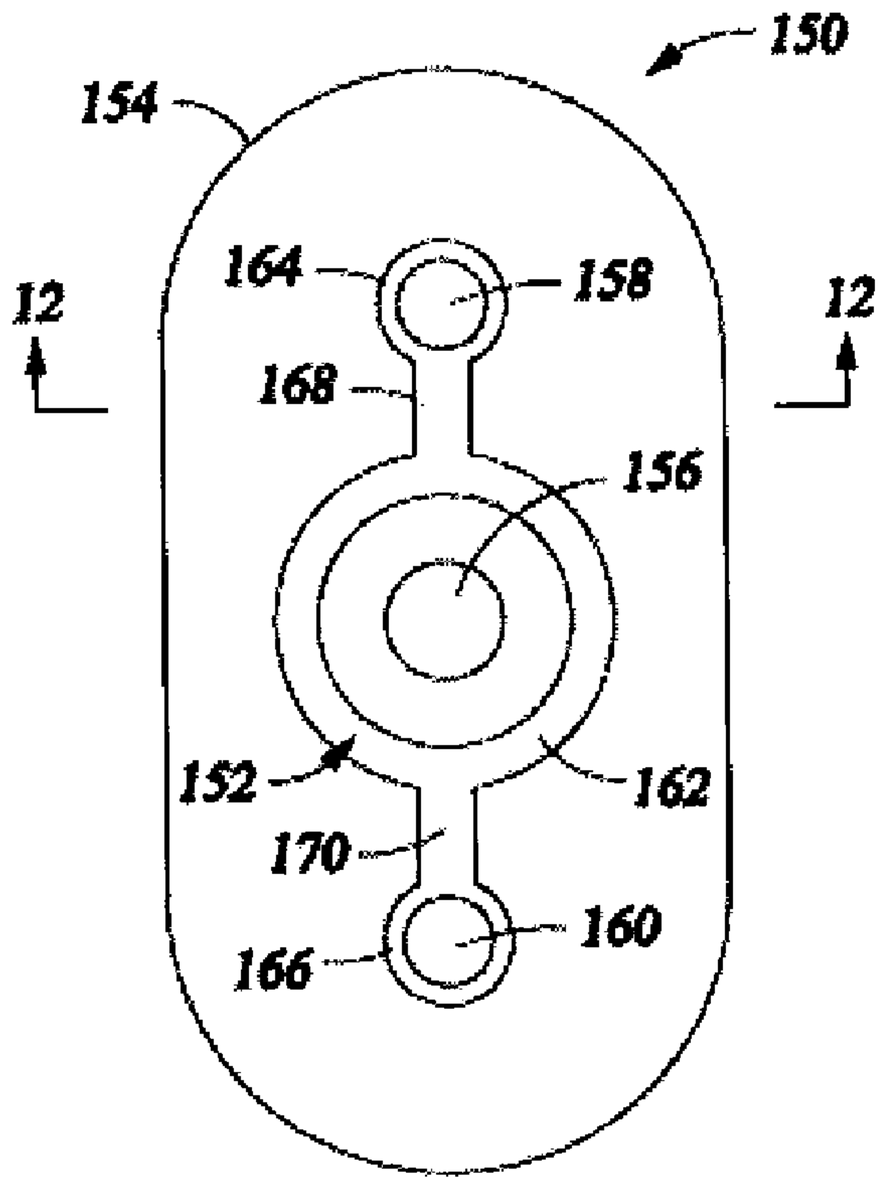


Fig. 11

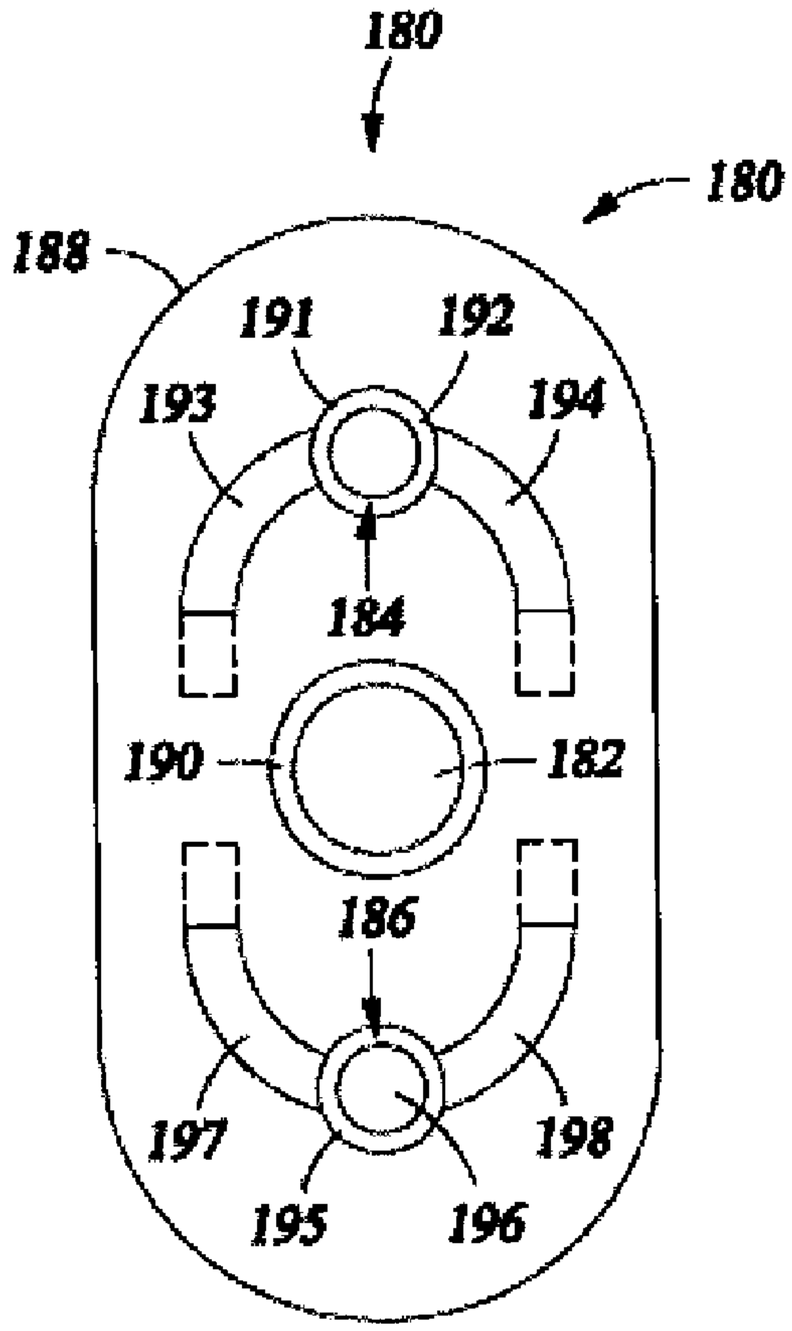


Fig. 13

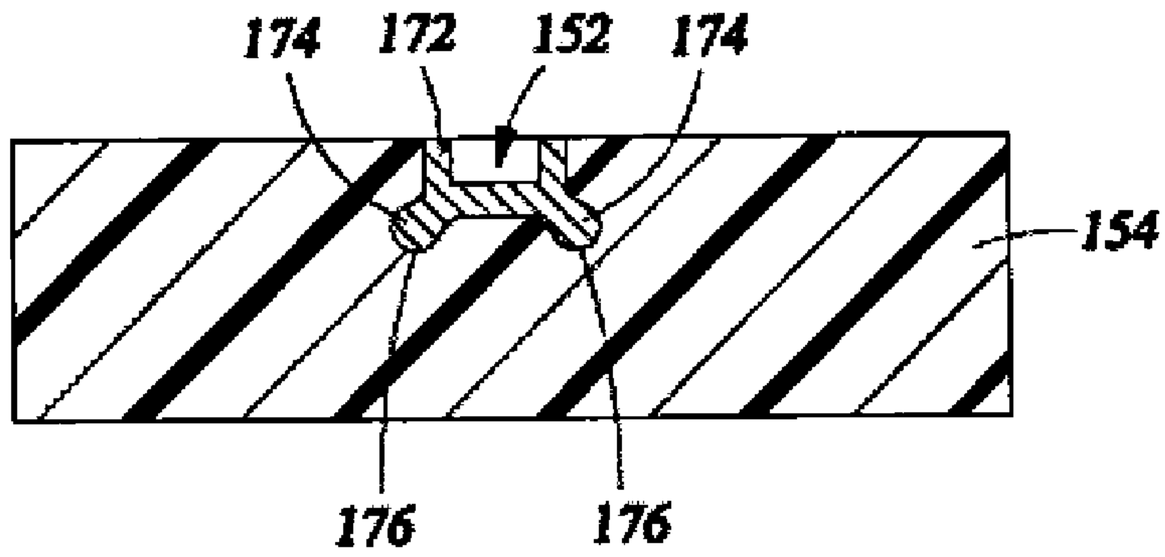


Fig. 12

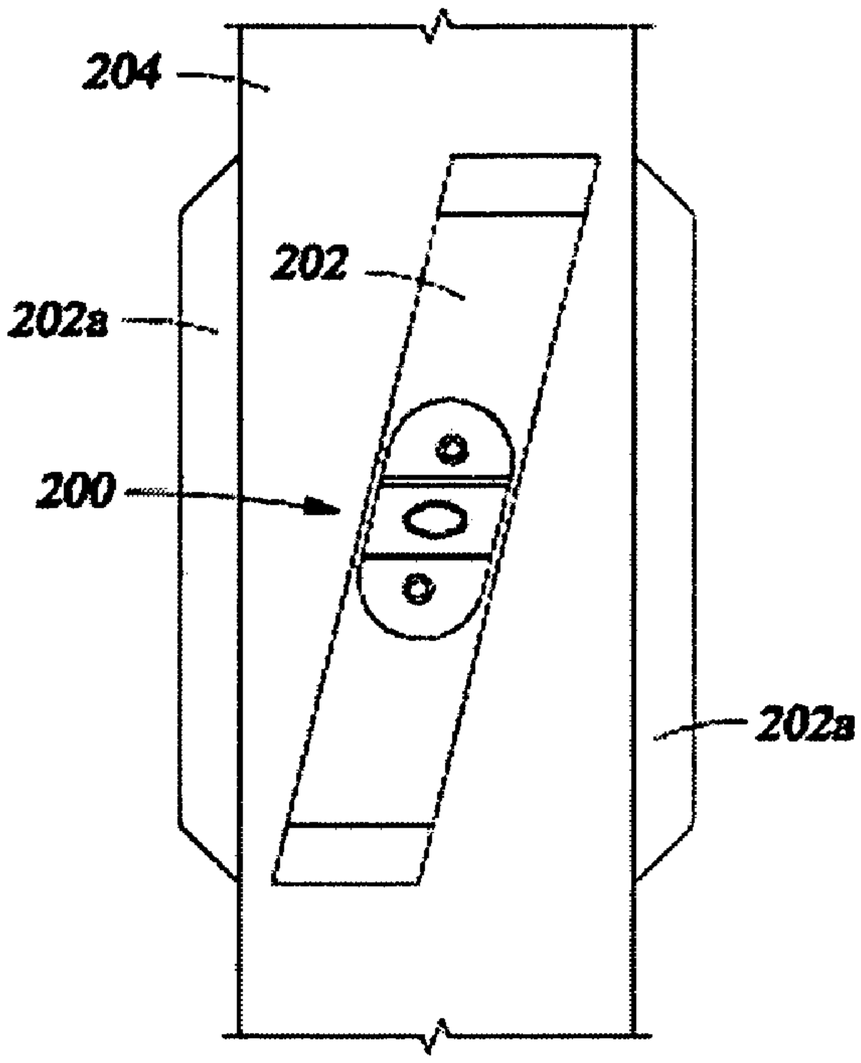


Fig. 14

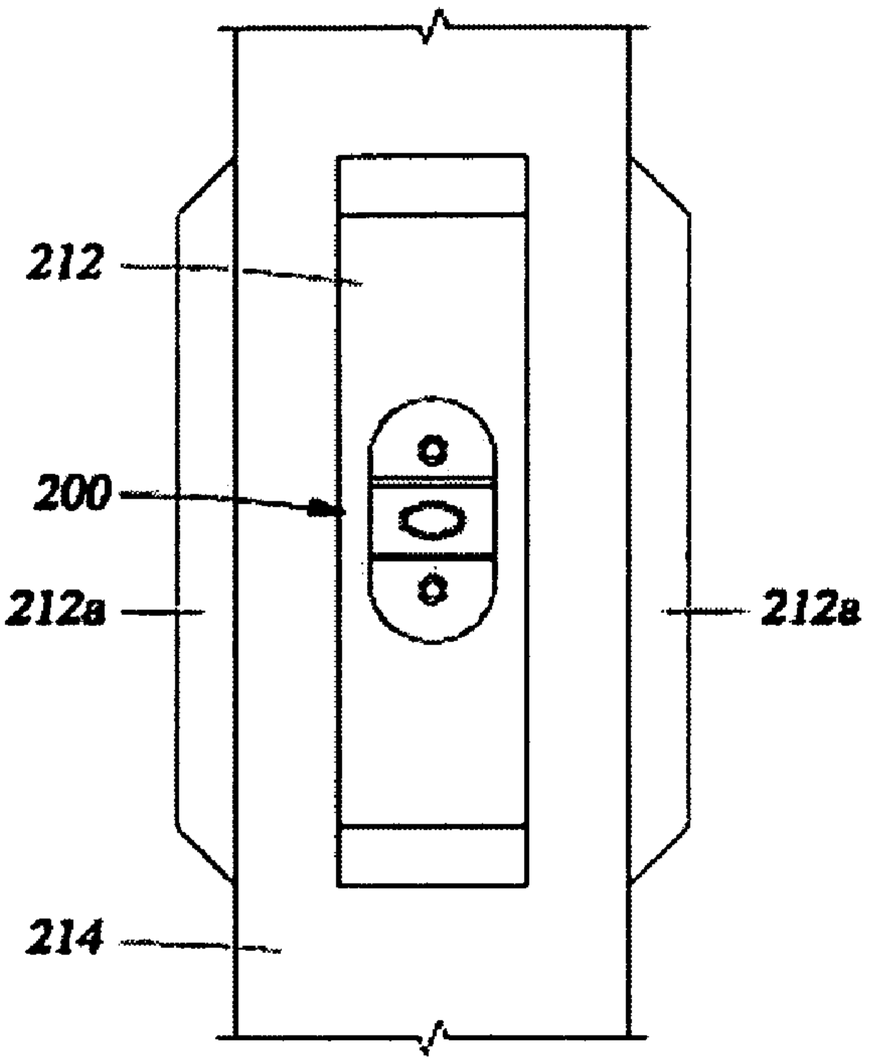
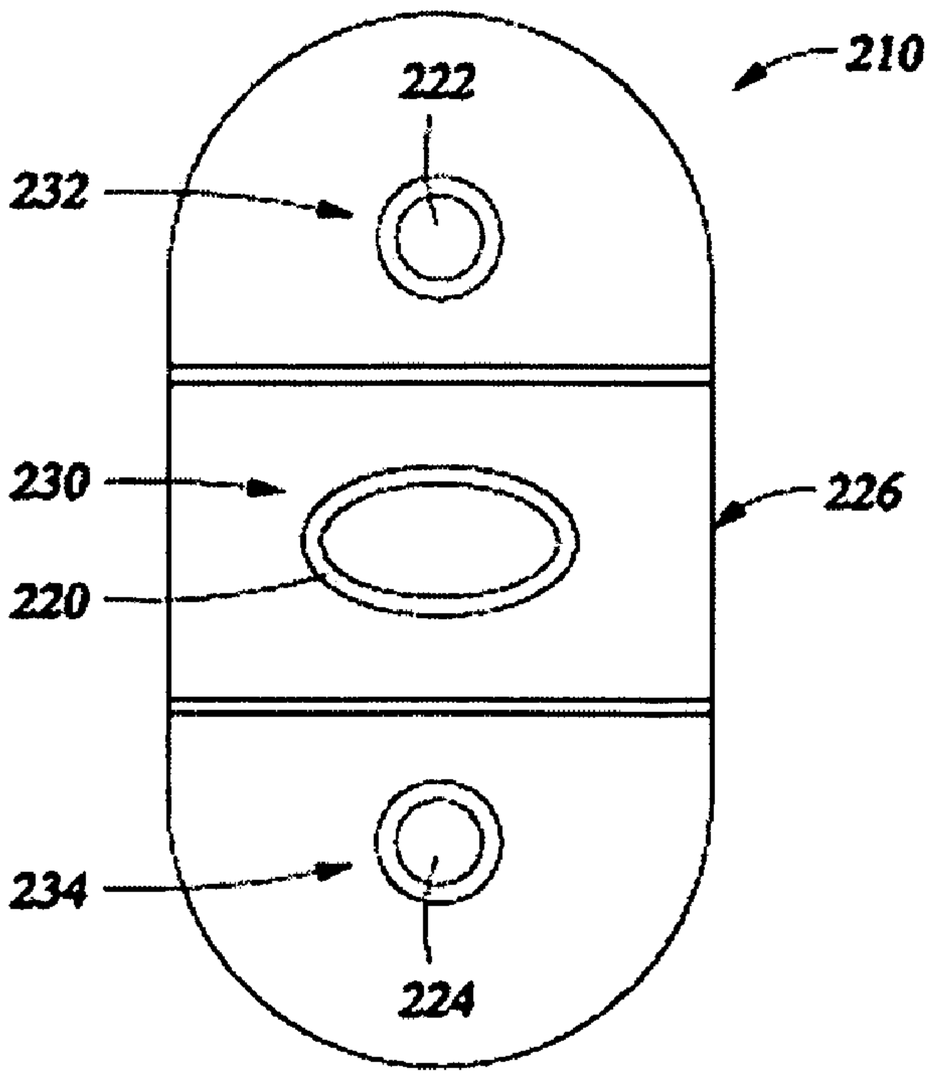


Fig. 15

Fig. 16



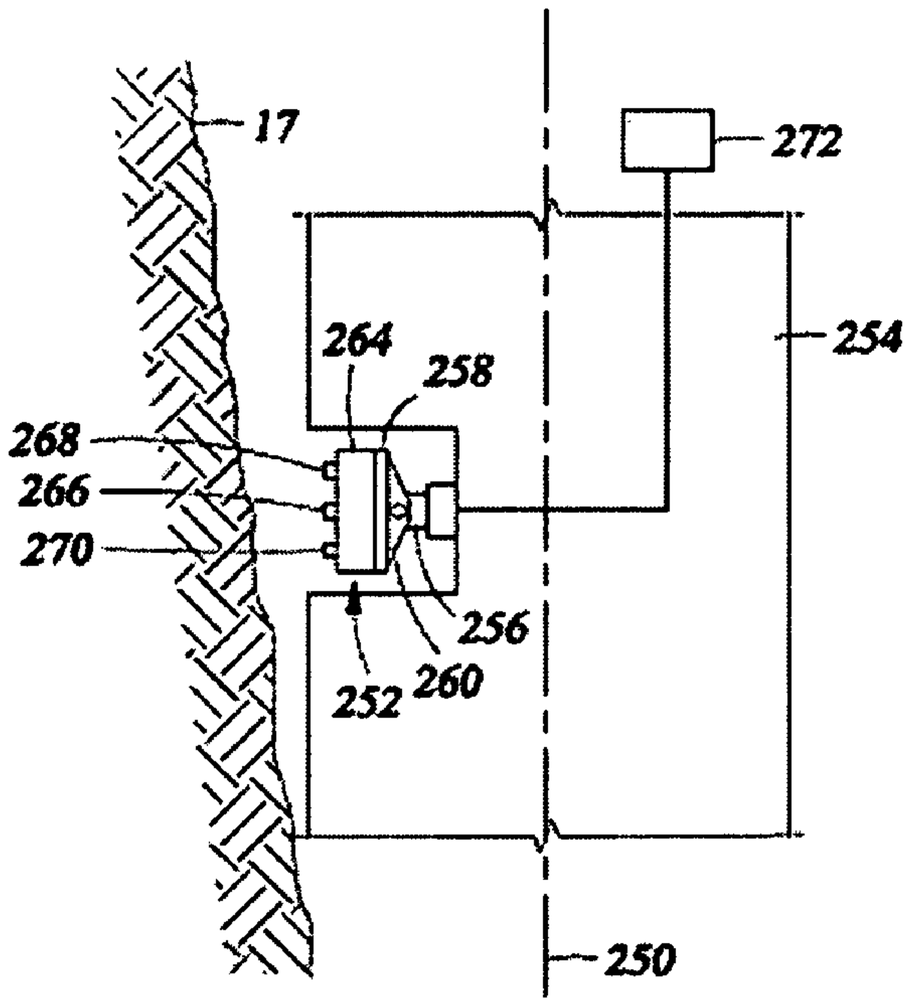


Fig. 17A

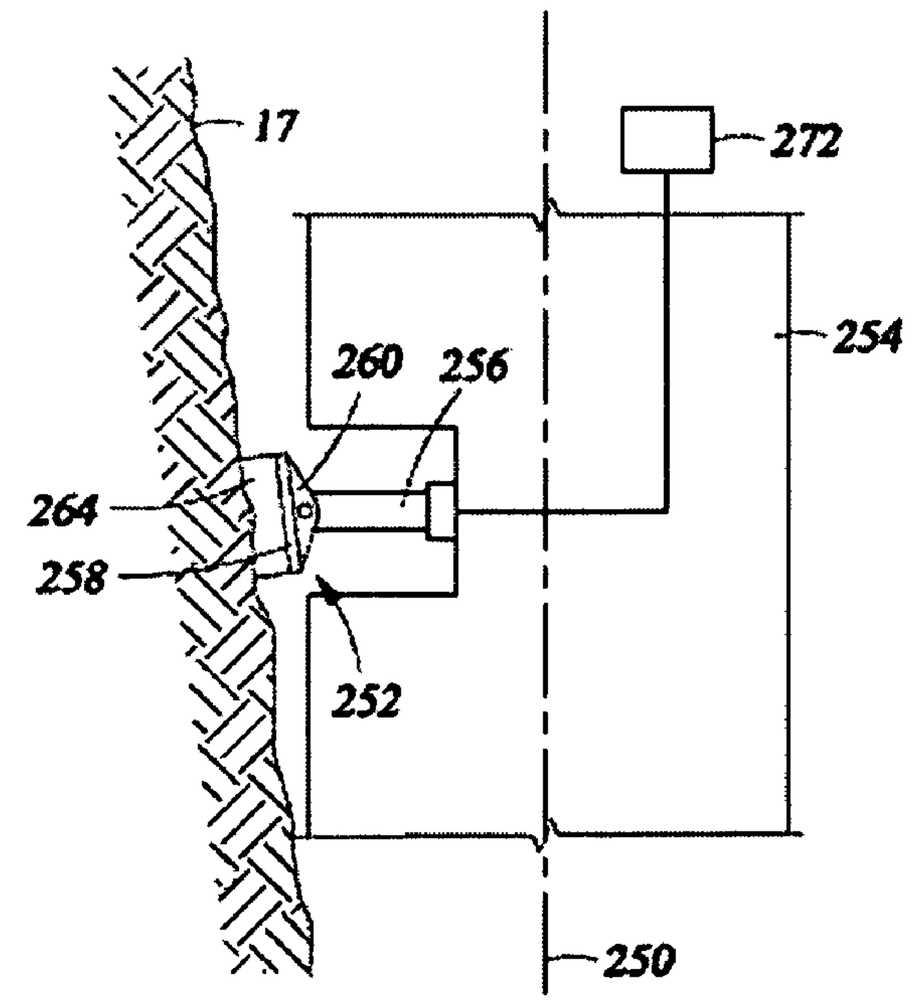


Fig. 17B

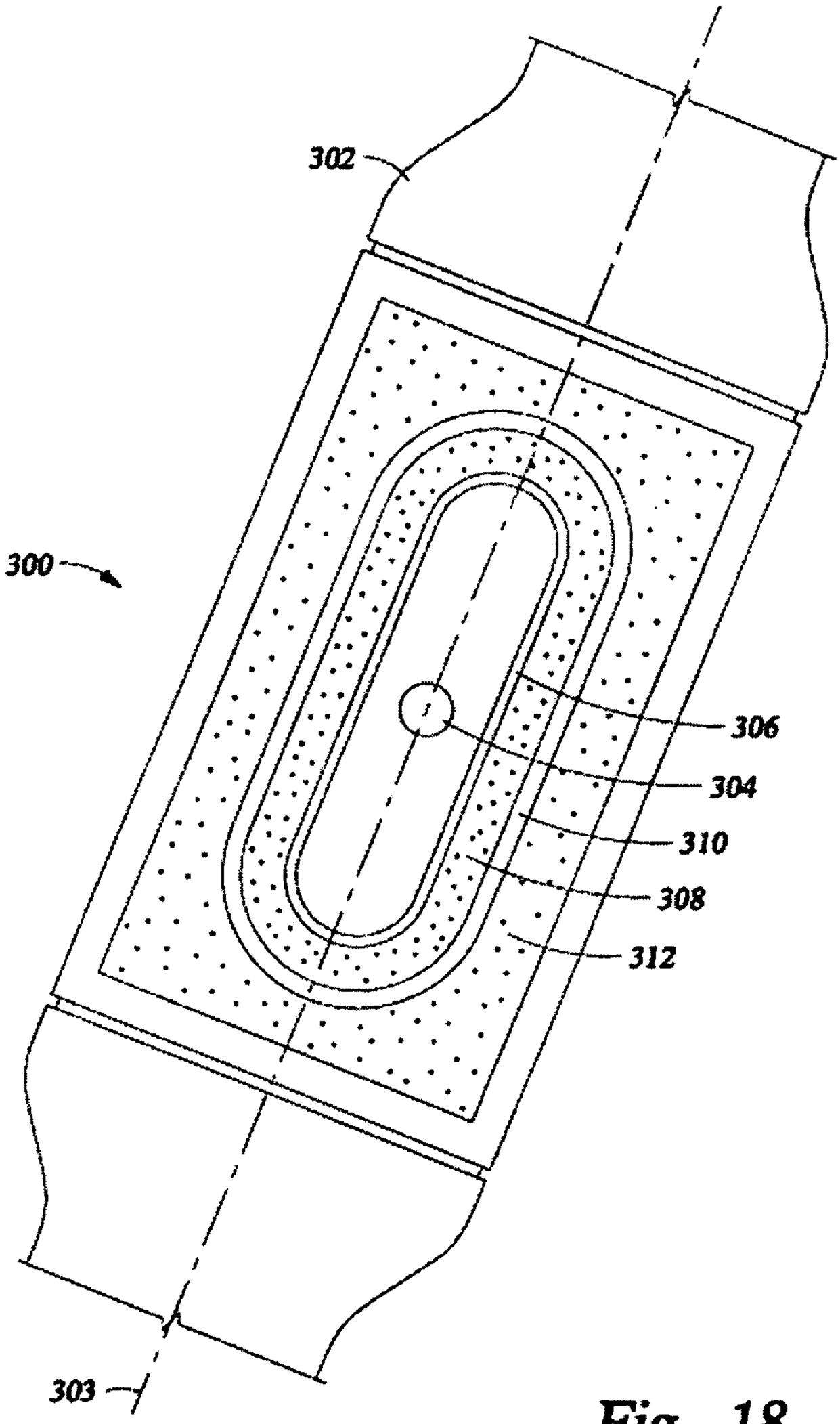


Fig. 18

FORMATION FLUID SAMPLING APPARATUS AND METHODS

BACKGROUND

1. Technical Field

This disclosure generally relates to investigations of subterranean formations, and more particularly to apparatus and methods for reducing the contamination of formation fluids drawn into a downhole formation testing and sampling tool.

2. Description of the Related Art

Wells are generally drilled into the ground or ocean bed to recover natural deposits of oil and gas, as well as other desirable materials that are trapped in geological formations in the Earth's crust. A well is typically drilled using a drill bit attached to the lower end of a "drill string" Drilling fluid, or "mud," is typically pumped down through the drill string to the drill bit. The drilling fluid lubricates and cools the drill bit, and it carries drill cuttings back to the surface in the annulus between the drill string and the wellbore wall.

For successful oil and gas exploration, it is necessary to have information about the subsurface formations that are penetrated by a wellbore. For example, one aspect of standard formation evaluation relates to the measurements of the formation pressure and formation permeability. These measurements are essential to predicting the production capacity and production lifetime of a subsurface formation.

One technique for measuring formation and fluid properties includes lowering a "wireline" tool into the well to measure formation properties. A wireline tool is a measurement tool that is suspended from a wireline in electrical communication with a control system disposed on the surface. The tool is lowered into a well so that it can measure formation properties at desired depths. A typical wireline tool may include a probe that may be pressed against the wellbore wall to establish fluid communication with the formation. This type of wireline tool is often called a "formation tester." Using the probe, a formation tester measures the pressure of the formation fluids, generates a pressure pulse, which is used to determine the formation permeability. The formation tester tool also typically withdraws a sample of the formation fluid that is either subsequently transported to the surface fox analysis or analyzed downhole.

In order to use any wireline tool, whether the tool be a resistivity, porosity or formation testing tool, the drill string must be removed from the well so that the tool can be lowered into the well. This is called a "trip" uphole. Further, the wireline tools must be lowered to the zone of interest, generally at or near the bottom of the hole. A combination of removing the drill string and lowering the wireline tools downhole are time-consuming measures and can take up to several hours, depending upon the depth of the wellbore. Because of the great expense and rig time required to "trip" the drill pipe and lower the wireline tools down the wellbore, wireline tools are generally used only when the information is absolutely needed or when the drill string is tripped for another reason, such as changing the drill bit. Examples of wireline formation testers are described, for example, in U.S. Pat. Nos. 3,934,468; 4,860,581; 4,893,505; 4,936,139; and 5,622,223

To avoid or minimize the downtime associated with tripping the drill string, another technique for measuring formation properties has been developed in which tools and devices are positioned near the drill bit in a drilling system. Thus, formation measurements are made during the drilling process and the terminology generally used in the art is "MWD" (measurement-while-drilling) and "LWD" (logging-while-drilling). A variety of downhole MWD and LWD drilling tools are commercially available.

MWD typically refers to measuring the drill bit trajectory as well as wellbore temperature and pressure, while LWD refers to measuring formation parameters or properties, such as resistivity, porosity, permeability, and sonic velocity, among others. Real-time data, such as the formation pressure, allows the drilling company to make decisions about drilling mud weight and composition, as well as decisions about drilling rate and weight-on-bit, during the drilling process. While LWD and MWD have different meanings to those of ordinary skill in the art, that distinction is not germane to this disclosure, and therefore this disclosure does not distinguish between the two terms.

Formation evaluation, whether during a wireline operation or while drilling, often requires that fluid from the formation be drawn into a downhole tool for testing and/or sampling. Various sampling devices, typically referred to as probes, are extended from the downhole tool. to establish fluid communication with the formation surrounding the wellbore and to draw fluid into the downhole tool. A typical probe is a circular element extended from the downhole tool and positioned against the sidewall of the wellbore. A rubber packer at the end of the probe is used to create a seal with the wellbore sidewall. Another device used to form a seal with the wellbore sidewall is referred to as a dual packer. With a dual packer, two elastomeric rings expand radially about the tool to isolate a portion of the wellbore therebetween. The rings form a seal with the wellbore wall and permit fluid to be drawn into the isolated portion of the wellbore and into an inlet in the downhole tool.

The mudcake lining the wellbore is often useful in assisting the probe and/or dual packers in making the seal with the wellbore wall. Once the seal is made, fluid from the formation is drawn into the downhole tool through an inlet by lowering the pressure in the downhole tool. Examples of probes and/or packers used in downhole tools are described in U.S. Pat. Nos. 6,301,959; 4,860,581; 4,936,139; 6,585,045; 6,609,568 and 6,719,049 and U. S. Patent Application Publication No. 2004/0000433.

Reservoir evaluation can be performed on fluids drawn into the downhole tool while the tool remains downhole. Techniques currently exist for performing various measurements, pretests and/or sample collection of fluids that enter the downhole tool. However, it has been discovered that when the formation fluid passes into the downhole tool, various contaminants, such as wellbore fluids and/or drilling mud primarily in the form of mud filtrate from the "invaded zone" of the formation, may enter the tool with the formation fluids. The invaded zone is the portion of the formation radially beyond the mudcake layer lining the wellbore where mud filtrate has penetrated the formation leaving the mudcake layer behind. These mud filtrate contaminates may affect the quality of measurements and/or samples of the formation fluids. Moreover, contamination may cause costly delays in the wellbore operations by requiring additional time for obtaining test results and/or samples representative of the formation fluid. Additionally, such problems may yield false results that are erroneous and/or unusable. Thus, it is desirable that the formation fluid entering into the downhole tool be sufficiently 'clean' or 'virgin' for valid testing. In other words, the formation fluid should have little or no contamination.

Attempts have been made to eliminate contaminates from entering the downhole tool with the formation fluid. For example, as depicted in U.S. Pat. No. 4,951,749, filters have been positioned in probes to block contaminates from entering the downhole tool with the formation fluid. Additionally,

as shown in U.S. Pat. No. 6,301,959, a probe is provided with a guard ring to divert contaminated fluids away from clean fluid as it enters the probe. More recently, U.S. Patent Application Publication No. 2006/0042793 discloses a central sample probe with an annular "guard" probe extending about an outer periphery of the sample probe, in an effort to divert contaminated fluids away from the sample probe.

Despite the existence of techniques for performing formation evaluation and for attempting to deal with contamination, there remains a need to manipulate the flow of fluids through the downhole tool to reduce contamination as it enters and/or passes through the downhole tool. It is desirable that such techniques are capable of diverting contaminants away from clean fluid.

Additionally, in while-drilling applications, the measuring apparatus is exposed to the extreme forces present during drilling operations. Any apparatus extending transversely through the wall of a drill string structure, such as a probe, will also weaken that structure. Thus, it is desirable to design probe apparatus so that it not only minimizes and/or withstands the while-drilling forces, but also minimizes any structural weaknesses in the drill string caused by the presence of the probe apparatus.

SUMMARY OF THE DISCLOSURE

A fluid sampling system is provided for retrieving a formation fluid sample from a formation surrounding a wellbore extending along a wellbore axis, the formation having a virgin fluid and a contaminated fluid therein. The system includes a sample inlet, a first guard inlet positioned adjacent to the sample inlet and spaced from the sample inlet in a first direction along the wellbore axis, and a second guard inlet positioned adjacent to the sample inlet and spaced from the sample inlet in a second, opposite direction along the wellbore axis. At least one cleanup flowline is fluidly connected to the first and second guard inlets for passing contaminated fluid, and an evaluation flowline is fluidly connected to the sample inlet for collecting virgin fluid.

In a refinement, the sample inlet is provided on a sample probe assembly including a sample inlet extension mechanism, the first guard inlet is provided on a first guard probe assembly including a first guard inlet extension mechanism, and the second guard inlet is provided on a second guard probe assembly including a second guard inlet extension mechanism, wherein the sample inlet, first guard inlet, and second guard inlet extension mechanisms are operable independently of one another.

In a related refinement, the sample probe assembly includes a sample inlet packer completely surrounding an outer periphery of the sample inlet, the first guard probe assembly includes a first guard inlet packer completely surrounding an outer periphery of the first guard inlet, and the second guard probe assembly includes a second guard inlet packer completely surrounding an outer periphery of the second guard inlet.

In a further refinement, the sample inlet packer, first guard inlet packer, and second guard inlet packer are formed as segments of a composite packer having a substantially contiguous outer periphery.

In a refinement, the sample probe assembly, first guard probe assembly, and second guard probe assembly are provided on a stabilizing blade of a drilling tool.

In yet another refinement, the sample inlet, first guard inlet, and second guard inlet are integrally provided on a single probe assembly including an inlet extension mechanism.

In still another refinement, the inlet packer includes a first packer segment disposed between the sample inlet and the first guard inlet and a second packer segment disposed between the sample inlet and the second guard inlet.

In a related refinement, the first and second packer segments further comprise a reinforcement material.

In a refinement, an exterior face of the inlet packer includes a guard channel.

In a further refinement, the system is associated with a wireline tool.

In another refinement, the system is associated with a drilling tool.

A probe assembly is also disclosed for use with a fluid sampling system to retrieve a formation fluid sample from a formation surrounding a wellbore extending along a wellbore axis, the formation having a virgin fluid and a contaminated fluid therein. The probe assembly includes an inlet extension mechanism and a sample inlet coupled to the inlet extension mechanism. A first guard inlet is coupled to the inlet extension mechanism, the first guard inlet being positioned adjacent to the sample inlet and spaced from the sample inlet in a first direction parallel to the wellbore axis. A second guard inlet is coupled to the inlet extension mechanism, the second guard inlet being positioned adjacent to the sample inlet and spaced from the sample inlet in a second, opposite direction parallel to the wellbore axis. An inlet packer completely surrounds outer peripheries of the sample inlet, first guard inlet, and second guard inlet.

In a related refinement, the probe packer includes a first packer segment disposed between the sample probe and the first guard probe and a second packer segment disposed between the sample probe and the second guard probe, wherein the first and second packer segments further comprise a reinforcement material.

In a further refinement, an exterior face of the probe packer includes a guard channel.

In another refinement, the guard channel includes a central ring section completely surrounding an outer periphery of the sample probe, a first guard ring section completely surrounding an outer periphery of the first guard probe, a second guard ring section completely surrounding an outer periphery of the second guard probe, a first link section extending between the central ring section and the first guard ring section, and a second link section extending between the central ring section and the second guard ring section.

In yet another refinement, the guard channel includes a guard ring section completely surrounding an outer periphery of the first guard probe and at least a first wing section connected to and extending away from the guard ring section.

In still another refinement, the guard channel further includes a second wing section connecting to and extending away from the guard ring section.

In a refinement, a second guard channel is provided having a guard ring section completely surrounding an outer periphery of the second guard probe and at least a first wing section connected to and extending away from the guard ring section.

In a related refinement, the guard channel is defined by a channel insert coupled to the probe packer.

In a further refinement, the channel insert is mechanically coupled to the probe packer.

In yet another refinement, the sample inlet, first guard inlet, and second guard inlet are pivotably coupled to the inlet extension mechanism.

A downhole tool is disclosed that is connected to a drill string positioned in a wellbore penetrating a subterranean formation along a wellbore axis. The tool includes a drilling collar having at least one stabilizing blade defining a blade

axis, an inlet extension mechanism housed within the stabilizing blade, and a probe assembly coupled to the inlet extension mechanism. The probe assembly comprises a sample inlet having a mouth portion with a first profile dimension in a direction parallel to the blade axis and a second profile dimension in a direction perpendicular to the blade axis, in which the first profile dimension is greater than the second profile dimension. An inner packer completely surrounds an outer periphery of the sample inlet, a guard inlet extends completely around an outer periphery of the inner packer, and an outer packer completely surrounds an outer periphery of the guard inlet

In a refinement, the probe assembly is pivotably coupled to the inlet extension Mechanism.

In a further refinement, the mouth portion has a generally oval shape cross-sectional profile, with the first profile dimension comprising a major axis and the second profile dimension comprising a minor axis.

In yet another refinement, the second profile dimension is less than approximately 3.5 inches.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more complete understanding of the disclosed methods and apparatuses, reference should be made to the embodiment illustrated in greater detail on the accompanying drawings, wherein:

FIG. **1** is a schematic view, partially in cross-section, of a downhole tool with a probe assembly according to the present disclosure, in which the downhole tool is a downhole drilling tool;

FIG. **2** is a schematic view, partially in cross-section, of a downhole tool with a probe assembly according to the present disclosure, in which the downhole tool is a wireline tool;

FIG. **3** illustrates one embodiment of a formation fluid sampling system made in accordance with this disclosure;

FIG. **4** is a schematic sectional view of the formation fluid sampling system of FIG. **3**;

FIGS. **5** and **6** schematically illustrate alternative probe arrangements for a formation fluid sampling system similar to that of FIG. **3**;

FIG. **7** illustrates an alternative formation fluid sampling systems;

FIG. **8** schematically illustrates fluid flow during use of the formation fluid sampling system of FIG. **7**;

FIG. **9** illustrates a further alternative formation fluid sampling system;

FIG. **10** is a detailed view of a packer employed in the formation fluid sampling system of FIG. **9**;

FIG. **11** is a plan view of yet another embodiment of a formation fluid sampling system made in accordance with this disclosure;

FIG. **12** is a cross-sectional view of the formation fluid sampling system taken along line A-A of FIG. **11**;

FIG. **13** is a plan view of still another embodiment of a formation fluid sampling system made in accordance with this disclosure;

FIG. **14** is a schematic illustration of the formation fluid sampling system housed in an angled stabilizing blade of a drill collar;

FIG. **15** is a schematic illustration of an alternative formation fluid sampling system similar to that of FIG. **14** housed in a vertical stabilizing blade of a drill collar;

FIG. **16** is an enlarged plan view of the formation fluid sampling system of FIG. **15**;

FIGS. **17**A and **17**B are schematic illustrations of a formation fluid sampling system having a pivotable probe assembly, made in accordance with this disclosure; and

FIG. **18** is a schematic illustration of yet another embodiment of probe assembly, in which the inlet is elongated for use on a stabilizing blade of a drill collar.

It should be understood that the drawings are not necessarily to scale and that the disclosed embodiments we sometimes illustrated diagrammatically and in partial views. In certain instances, details which are not necessary for an understanding of the disclosed methods and apparatuses or which render other details difficult to perceive may have been omitted. It should be understood, of course, that this disclosure is not limited to the particular embodiments illustrated herein.

DETAILED DESCRIPTION

This disclosure relates to probe assemblies and configurations described below that may be used with a downhole tool, either in a drilling environment or in a wireline environment. The apparatus and methods disclosed herein reduce the contamination of formation fluid samples. In some refinements, this disclosure relates to the relative positioning of multiple, independently operable probe assemblies. In one or more other refinements, a fluid sampling system includes a single assembly having multiple probes. In addition, a probe configuration particularly suited to while-drilling applications is disclosed.

The phrase "formation evaluation while drilling" refers to various sampling and testing operations that may be performed during the drilling process, such as sample collection, fluid pump out, pretests, pressure tests, fluid analysis, and resistivity tests, among others. It is noted that "formation evaluation while drilling" does not necessarily mean that the measurements are made while the drill bit is actually cutting through the formation. For example, sample collection and pump out are usually performed during brief stops in the drilling process. That is, the rotation of the drill bit is briefly stopped so that the measurements may be made. Drilling may continue once the measurements are made. Even in embodiments where measurements are only made after drilling is stopped, the measurements may still be made without having to trip the drill string.

In the exemplary embodiments, a probe assembly according to the present disclosure is carried by a downhole tool, such as the drilling tool **10** of FIG. **1** or the wireline tool **10'** of FIG. **2**. The probe assembly may also be used in other downhole tools adapted to draw fluid therein, such as coiled tubing, casing drilling, and other variations of downhole tools.

FIG. **1** depicts a downhole drilling tool **10** deployed from a rig **5** and advanced into the earth to form a wellbore **14**. The wellbore penetrates a subterranean formation F containing a formation fluid **21**. The downhole drilling tool is suspended from the drilling rig by one or more drill collars **11** that form a drill string **28**. "Mud" is pumped through the drill string **28** and out bit **30** of the drilling tool **10**. The mud is pumped back up through the wellbore and to the surface for filtering and recirculation. As the mud passes through the wellbore, it forms a mud layer or mudcake **15** along the wellbore wall **17**. A portion of the mud infiltrates the formation to form an invaded zone **25** of the formation F.

In the illustrated embodiment, the drilling tool **10** is provided with a probe **26** for establishing fluid communication with the formation F and drawing the fluid **21** into the downhole tool, as indicated by the arrows. As shown in FIG. **1**, the probe is positioned in a stabilizer blade **23** of the drilling tool and extended therefrom to engage the wellbore wall. The

stabilizer blade **23** comprises one or more blades that are in contact with the wellbore wall to limit "wobble" of the drill bit **30**. "Wobble" is the tendency of the drill string, as it rotates, to deviate from the axis of the wellbore **14** and cause the drill bit to change direction. Advantageously, a stabilizer blade **23** is already in contact with the wellbore wall, thus requiring less extension of a probe to establish fluid communication with the formation fluids if the probe is disposed in the stabilizer blade **23**.

Fluid drawn into the downhole tool using the probe **26** may be measured to determine, for example, pretest and/or pressure parameters. Additionally, the downhole tool may be provided with devices, such as sample chambers, for collecting fluid samples for retrieval at the surface. Backup pistons **8** may also be provided to assist in applying force to push the drilling tool and/or probe against the wellbore wall. The drilling tool may be of a variety of chilling tools, such as Measurement-While-Drilling ("MWD"), Logging-While-Drilling ("LWD"), casing drilling, or other system. An example of a drilling tool usable for performing various downhole tests is depicted in U.S. patent application Ser. No. 10/707,152, filed on Nov. 24, 2003, the entire contents of which are hereby incorporated by reference.

The downhole drilling tool **10** may be removed from the wellbore and a wireline tool **10'** (FIG. **2**) may be lowered into the wellbore via a wireline cable **18**. An example of a wireline tool capable of sampling and/or testing is depicted in U.S. Pat. Nos. 4,936,139 and 4,860,581, the entire contents of which are hereby incorporated by reference. The downhole tool **10'** is deployable into borehole **14** and suspended therein with a conventional wireline **18**, or conductor or conventional tubing or coiled tubing, below the rig **5**. The illustrated tool **10'** is provided with various modules and/or components **12** including, but not limited to, a probe **26'** for establishing fluid communication with the formation F and drawing the fluid **21** into the downhole tool as shown by the arrows. Backup pistons **8** may be provided to further thrust the downhole tool against the wellbore wall and assist the probe in engaging the wellbore wall The tools of FIGS. **1** and **2** may be modular as shown in FIG. **2** or unitary as shown in FIG. **1**, or combinations thereof.

Turning to FIG. **3**, a probe assembly **30** is recessed within a stabilizing blade **32** of a drill collar **34**. The probe assembly **30** includes a sample inlet **36**, a first guard inlet **38**, and a second guard inlet **40**. Each of the inlets **36**, **38**, **40** is oriented generally transversely to a longitudinal axis of the drill collar **34** and is normally in a retracted position so that the inlets **36**, **38**, **40** are housed within one or more cavities formed in the stabilizing blade **32**. A dedicated probe extension mechanism, such as a hydraulic arm as described in U.S. Pat. Nos. 6,230,557; 4,860,581; and 4,936,139 commonly assigned to the assignee of the present application, the entire contents of which are hereby incorporated by reference, is operatively coupled to each inlet **36**, **38**, **40** to selectively and independently move the associated inlet to an extended position. In the extended position, the inlets **36**, **38**, or **40** may extend outside of the cavity to place the inlet in better position to contact the wellbore wall **17**. Back up pistons **42***a-c* are extendible to move the probe assembly **30** toward the formation F.

While the exemplary embodiment describes inlets that are extendable, it will be appreciated that the inlets may be non-extendable and therefore fixed with respect to the position of the drill collar **34**. In addition, the probe assembly **30** may include a protector which provides mechanical protection to the inlets during drilling and/or tripping operations and which provides mechanical protection to the mudcake against erosion generated by flowing mud. One such protector is described in U.S. Pat. No. 6,729,399 commonly assigned to the assignee of the present application, the entire contents of which are hereby incorporated by reference.

As shown in FIG. **4**, fluid flowlines are connected to the inlets for passing either waste fluid or clean fluid. In the illustrated embodiment, sample inlet **36** is fluidly connected to an evaluation flowline **52** by an inlet flowline **54***a*. A bypass flowline **56***a* fluidly communicates between the sample probe **38** and a clean up flowline **58**. The first guard inlet **38** is also fluidly connected to the evaluation and clean up flowlines **52**, **58** by an inlet flowline **54***b* and a bypass flowline **56***b*, respectively. Similarly, the second guard inlet **40** is in fluid communication with the evaluation and clean up flowlines **52**, **58** by an inlet flowline **54***c* and a bypass flowline **56***c*. Valves **60***a-f* are provided in the inlet and bypass flowlines **54**, **56** to direct fluid flow to the evaluation and clean up flowlines **52**, **58**, as desired. Fluid sensors, such as optical fluid analyzers **46***a*, **46***b*, are associated with the flowlines **52**, **58** to provide feedback regarding characteristics or other information regarding the fluid passing through the flowlines.

A pump **62** is fluidly coupled to the evaluation and clean up flowlines **52**, **58**. A sample storage assembly (not shown) may fluidly communicate with the evaluation flowline **52** upstream of the point where the evaluation flowline **52** and clean up flowline **58** are connected, to provide means for collecting a clean fluid sample. A pump discharge flowline **64** may communicate between the pump and the wellbore **14** for discharging contaminated formation fluid. The pump **62** and valves **60***a-f* may be operated in various manners to clear contaminated formation fluid from the immediate area of the probes **36**, **38**, **40** and to draw clean formation fluid into the evaluation flowline **52**, such as the methods disclosed in U.S. Patent Application Publication No. 2006-0042793, the entire contents of which are hereby incorporated by Reference.

Each of the inlets **36**, **38**, **40** of the probe assembly **30** includes a packer for sealing with the wellbore wall **17**. As illustrated in FIGS. **3** and **4**, a sample inlet packer **80** is provided that completely surrounds an outer periphery of the sample inlet **36**. Similarly, first and second guard inlet packers **82**, **84** completely surround outer peripheries of the first and second guard inlets **38**, **40**, respectively.

The inlets **36**, **38**, **40** are positioned relative to one another to reduce the amount of contaminants that reach the sample inlet **36**. In the illustrated embodiment, the first guard inlet **38** is positioned adjacent to and above the sample inlet **36** while the second guard inlet **40** is positioned adjacent to and below the sample inlet **36**. This arrangement of inlets minimizes or prevents fluid from the invaded zone from entering the sample inlet **36**. The invaded zone **25** is the area where mud filtrate has entered the formation F radially from the wellbore **14**, leaving a layer of mudcake lining the wellbore wall **17**. Once filtrate-laden formation fluid from the invaded zone has been removed from the circumferential area surrounding the inlets **36**, **38**, **40**, the first and second guard inlets **38**, **40** prevent mud filtrate and contaminated fluid from migrating axially toward the sample inlet **36**. As a result, the sample inlet **36** retrieves formation fluid having little or no filtrate contamination.

The distance between the inlets **36**, **38**, **40** must balance performance and structural considerations. On the one hand, it is desirable to locate the inlets **36**, **38**, **40** as close to one another as possible, thereby to minimize the volume of fluid that must be initially pumped from the formation before a clean fluid flow is obtained at the sample inlet **36**. On the other hand, each inlet **36**, **38**, **40** requires an aperture to be formed through an exterior of the drilling tool. In while-drilling applications, the drill collar carrying the probe assembly must be

structurally sound to withstand the forces experienced during drilling operations. In addition, farther spaced inlets **36**, **38**, **40** reduce the chance of cross-contamination of flow streams into each inlet. As a practical matter, therefore, it is preferable to have a space between each adjacent pail of inlets of at least one inlet diameter.

Various alternative inlet configurations and combinations may be used without departing from the scope of this disclosure. For example, instead of providing vertically aligned inlets as shown in FIGS. **3** and **4**, the sample inlet **36** may be azimuthally offset from the first and second guard inlets **38**, **40**, as shown in FIG. **5**. In this embodiment, the sample inlet **36** extends from a first side of the drill collar **11** while the first and second guard inlets **38**, **40** extend from a second, opposite side of the drill collar **11**. This configuration is still effective to prevent filtrate from reaching the sample inlet **36** because the first and second guard inlets **38**, **40** remove fluid from an area of the formation lying within an annular band surrounding each inlet Alternatively, an additional guard inlet **86** may be provided as shown in FIG. **6**.

An alternative probe assembly embodiment having multiple inlets actuated by a single extension mechanism is illustrated in FIGS. **7** and **8**. A probe assembly **100** is illustrated as recessed within a stabilizer blade **101** of a drill collar **103**. The probe assembly **100** includes sample inlet **102**, a first guard inlet **104**, and a second guard inlet **106**. The inlets **102**, **104**, **106** may be operatively coupled to a single extension mechanism that simultaneously advances and retracts the probes or, alternatively, the inlets may be non-extendable. The probe assembly **100** further includes a single packer **110** that completely surrounds outer peripheries of the sample inlet **102**, first guard inlet **104**, and second guard inlet **106**. The inlets **102**, **104**, **106** are generally vertically aligned with the sample inlet **102** positioned in between the first and second guard inlets **104**, **106**. A back up piston **107** is provided for positioning the assembly **100** adjacent the wellbore wall **17**.

In operation, the drill collar **103** carrying the probe assembly **100** is positioned within the wellbore **14**, as illustrated in FIG. **8**. To perform testing, the probe assembly **100** is positioned adjacent the wellbore wall **17**, either by extending the inlets **102**, **104**, **106** away from the drill collar **103** or by extending the back up piston **107**, or both, until the packer **110** contacts the wellbore wall **17** and forms a seal with the mudcake **15**. As discussed above, the drilling mud seeps into the formation through the wellbore wall **17** and creates an invaded zone **25** about the wellbore **14**, leaving a layer of mudcake **15** that lines the wellbore wall **17**. The invaded zone **25** contains mud and other wellbore fluids that contaminate the surrounding formation, including the formation F having a zone of clean formation fluid **114** contained therein. As illustrated in FIG. **8**, operation of the probe assembly **100** will remove contaminated formation fluid from the area immediately surrounding the inlets **102**, **104**, **106**. During operation, filtrate may continue to migrate axially through the invaded zone **25**, in either the upward or downward direction. Any such migrant filtrate will be removed by the first and second guard inlets **104**, **106** prior to reaching the sample inlet **102**, thereby allowing the sample inlet **102** to retrieve substantially clean formation fluid samples.

FIGS. **9** and **10** illustrate an alternative embodiment of a single probe assembly having multiple inlets. A probe assembly **120** is shown coupled to a drill collar **122**. The probe assembly **120** includes a sample inlet **124**, a first guard inlet **126**, and a second guard inlet **128**. A single packer **130** is provided having an outer portion **132** surrounding the exterior portions of the sample inlet **124**, first guard inlet **126**, and second guard inlet **128**. The packer **130** also includes a first interior segment **134** extending between the sample inlet **124** and the first guard inlet **126**, and a second interior segment **136** extending between the sample inlet **124** and the second guard inlet **128**. In the illustrated embodiment, the exterior peripheries of the inlets **124**, **126**, **128** trace an oval shape that is interrupted by the first and second packer segments **134**, **136**. In this arrangement, the inlets **124**, **126**, **128** are positioned more closely to one another in the vertical direction, which may improve the clarity of the formation fluid sample retrieved through the sample probe **124**.

The first and second packer segments **134**, **136** may be reinforced to improve their resistance to pressure differentials. A reinforcement material, such as a metal, composite, or other high strength material, may be molded into the first and second segments **134**, **136** of the rubber packer **130**. The first and second segments **134**, **136** prevent filtrate from migrating vertically into the sample inlet **124**. While the left and right side sections of the sample inlet **124** are left relatively unprotected, it has been found that the circumferential area surrounding the sample inlet **124** remains relatively clear of filtrate once it has been initially evacuated, and that the first and second guard inlets **126**, **128** prevent vertical migration into this area of the formation. Additionally, the sample inlet **124** configuration illustrated in FIGS. **9** and **10** allow these unprotected side sections to be fairly small, thereby further minimizing the potential for filtrate or formation fluid contaminated with filtrate to reach the sample inlet **124**. While the inlets **124**, **126**, **128** are shown with shapes that fit within an oval shaped packer outer portion **132**, it will be appreciated that other shapes may be used without departing from the scope of this Disclosure.

A further refinement is illustrated in FIGS. **11** and **12**, which show a probe assembly **150** with a guard channel **152** formed in an exterior face of a packer **154**. The probe assembly **150** includes a sample inlet **156**, a first guard inlet **158**, and a second guard inlet **160**. The packer **154** completely surrounds the outer peripheries of the inlets **156**, **158**, **160**. The guard channel **152** is formed as a recess in the exterior surface of the packer **154**. The guard channel **152** includes a central ring section **162** that is spaced from and completely surrounds an outer periphery of the sample inlet **156**, a first guard ring section **164** that borders on and completely surrounds an outer periphery of the first guard inlet **158**, and a second guard ring section **166** that borders on and completely surrounds an outer periphery of the second guard inlet **160**. A first link section **168** extends between the central ring section **162** and the first guard ring section **164**, and a second link section **170** extends between the central ring section **162** and the second guard ring section **166**.

In the illustrated embodiment, the guard channel **152** is formed in a channel insert **172** that is coupled to the packer **154**. For example, the channel insert **172** may be mechanically coupled to the packer **154** such as by forming tabs **174** that are received in anchor slots **176** to form a dove-tail like connection, as best shown in FIG. **12**. The channel insert **172** may be made from a low modulus material, such as titanium alloy, to better conform to the wall of the wellbore. It will be appreciated that low modulus materials other than titanium alloy may be used without departing from the scope of this disclosure. The channel may be defined by a structural conduit as shown in FIG. **12**, or may be defined by a porous material with integral flow passages.

An alternative assembly using a different guard channel configuration is illustrated in FIG. **13**. A guard probe assembly **180** includes a sample inlet **182**, a first guard inlet **184**, and a second guard inlet **186**. A packer **188** completely surrounds the outer peripheries of the sample, first guard, and second

guard inlets, **182**, **184**, **186**. A sample inlet channel **190** is provided on an exterior surface of the packer **188** that borders on and completely surrounds an outer periphery of the sample inlet **182**. A first guard channel **191** includes a first guard ring section **192** that borders on and completely surrounds an outer periphery of the first guard inlet **184**. First and second wings **193**, **194** fluidly communicate with the first guard ring section **192** and extend laterally outwardly from opposite sides of the first guarding section **192**. The first and second wing sections **193**, **194** are curved to extend toward the sample inlet **182**, as shown in FIG. **13**. A second guard channel **195** includes a second guard ring section **196** that borders on and completely surrounds an outer periphery of the second guard inlet **186**. The second guard channel **195** includes first and second wing sections **197**, **198** that fluidly communicate with and extend from opposite sides of the second guard ring section **196**. The first and second wings **197**, **198** are also curved to extend toward the sample inlet **182**.

Further alternative embodiments of a probe assembly are illustrated in FIGS. **14** and **15**. FIG. **14** illustrates a probe assembly **200** positioned on a probe/stabilizer blade **202** of a drill collar **204**, which also includes stabilizer blades **202***a*. The probe/stabilizer blade **202** is angled with respect to a vertical axis of the drill collar **204**. In FIG. **14**, a probe assembly **210** is shown coupled to a probe/stabilizer blade **212** of a drill collar **214**, wherein the probe/stabilizer blade **212** is substantially parallel to a vertical axis of the drill collar **214**. The drill collar **214** also includes additional stabilizer blades **212***a*.

The probe assembly **210** is illustrated in greater detail at FIG. **16**. The probe assembly **210** includes a sample inlet **220**, a first guard inlet **222**, and a second guard inlet **224**. Similar to previous embodiments, the inlets **220**, **222**, **224** are substantially vertically aligned, with the sample inlet **220** positioned between the first and second guard inlets **222**, **224**.

A composite packer **226** completely surrounds the outer peripheries of the sample inlet **220**, first guard inlet **222**, and second guard inlet **224**. The composite packer **226** may include segments that permit independent extension or retraction of each inlet **220**, **222**, **224**. In the illustrated embodiment, the composite packer **226** includes a sample inlet segment **230**, a first guard inlet segment **232**, and a second guard inlet segment **234**. To independently actuate each probe, a sample inlet extender is operatively coupled to the sample inlet **220**, a first guard inlet extender is operatively coupled to the first guard inlet **222**, and a second guard inlet extender is operatively coupled to the second guard inlet **224**. The segments **230**, **232**, **234** are shaped so that the composite packer **226** has a substantially contiguous outer periphery. In the illustrated embodiment, the outer periphery has an oval shape.

The sample inlet **220** may be shaped to maximize fluid withdrawal in a circumferential direction while minimizing fluid withdrawal from the formation in a vertical direction. In the illustrated embodiment, the sample inlet **220** has an oval shape with a major axis extending in a substantially horizontal direction and a minor axis extending in a substantially vertical direction, parallel to the wellbore axis. While an oval shape is illustrated, other shapes, including elongate and oblong profiles, may be used without departing from the scope of this disclosure.

FIGS. **17**A and **17**B illustrate an alternative embodiment of a sample probe assembly that is pivotable to conform to contour of the wellbore wall, thereby more reliably forming a seal therewith. It will be appreciated that the wellbore wall **17** is not always parallel to an axis **250** of a downhole tool. Consequently, the packer of a probe assembly may be presented at an angle to the wellbore, thereby reducing the ability to sufficiently seal with the wellbore wall. As shown in FIG. **16**A, a probe assembly **252** is coupled to a drill collar **254** by a probe extender **256**. The probe assembly **252** includes a backing plate **258** having a bracket **260** attached thereto. The bracket **260** is pivotably coupled to an end of the probe extender **256**. The backing plate **258** carries a packer **264**, a sample inlet **266**, a first guard inlet **268**, and a second guard inlet **270**. The probe extender **256** may be provided as an actuating cylinder that is operatively coupled to a power supply, such as a source of hydraulic fluid **272**.

In operation, the probe extender **256** may be actuated to move the probe assembly **252** from a retracted position where the assembly is spaced from the wellbore wall **17**, shown in FIG. **17**A, to an extended position where the assembly engages the wellbore wall **17**, shown in FIG. **17**B. The pivotable connection between the extender **256** and the backing plate **258** allows the packer **264** to tilt complementary to the wellbore wall **17**, thereby more reliably sealing with the wall.

FIG. **18** illustrates a further embodiment of a probe assembly **300** having an elongated profile to provide improved fluid flow while meeting the size constraints associated with use in a stabilizing blade **302** of a drilling tool, such as drilling collar **307**. The probe assembly **300** is housed within a cavity **309** formed in the blade **302** so that the assembly **300** may be recessed during drilling operations. An extension mechanism (not shown) is provided to extend the assembly **300** into contact with the wellbore wall to perform sampling operations.

The assembly **300** includes a sample inlet **304** having an expanded mouth portion **306**. The mouth portion **306** is elongated along a longitudinal axis **303** of the blade **302** to provide an enlarged communication surface for engaging the formation. More specifically, the mouth portion has a first profile dimension in a direction parallel to the blade axis **303** and a second profile dimension in a direction perpendicular to the blade axis **303**, in which the first profile dimension is greater than the second profile dimension. In the illustrated embodiment, the mouth portion has a generally oval shape cross-sectional profile, with the first profile dimension comprising a major axis and the second profile dimension comprising a minor axis. To meet the space restrictions presented by the blade stabilizer, the second profile dimension may be less than approximately 3.5 inches.

The sample inlet **304** is surrounded by an inner packer **308**. An oval-shaped guard inlet **310** completely surrounds the inner packer **308** and sample inlet **304**. The guard inlet **310** has a profile that is elongated along the longitudinal axis of the blade, similar to the sample inlet **304**. An outer packer **312** surrounds a periphery of the guard inlet **310**. The inner and outer packers **308**, **312** have a thickness and/or are formed of a material that provides sufficient strength to withstand the pressure differentials generated during operation of the probe assembly **300**.

The probe assembly **300** illustrated in FIG. **18** is particularly suited for use in a stabilizing blade **302** in while-drilling applications. As noted above, it is desirable to minimize the size of the inlets to maintain structural integrity of the drill collar. When provided within a stabilizing blade, inlet size is further restricted by the dimensions of the blade, particularly the relatively narrow width of the blade. As a result, the guard inlet must be reduced form a width of 4-10 inches or more (as is typical for wireline applications) to approximately 3.5 inches or less to fit within the stabilizing blade. This disclosure is not limited to these specific dimensions, as the size of the guard inlet may be commensurate with the overall dimensions of the wellbore or the tool in which the guard inlet

resides. After leaving sufficient room for the inner packer **308**, only a relatively narrow space is left for the sample inlet **304**. The sample inlet **304**, however, must have a communication area that engages the formation that is sufficiently large to ensure adequate liquid flow. The elongated, oval shape of the mouth portion **306** increases the communication area of the sample inlet **304** while meeting space restrictions imposed by the blade structure.

With the increased communication area provided by the mouth portion **306**, it can be more difficult to form a sufficient seal between the packers **308**, **312** and the formation, since the increased contact area is more likely to encounter ruggosity or other formation surface deviations. The pivotable probe head discussed above in connection with FIGS. **17**A and **17**B may be employed with the elongated profile to minimize the effects of formation surface irregularities.

While only certain embodiments have been set forth, alternatives and modifications will be apparent from the above description to those skilled in the art. These and other alternatives are considered equivalents and within the spirit and scope of this disclosure and the appended claims.

What is claimed:

1. A fluid sampling system for retrieving a formation fluid sample from a formation surrounding a wellbore extending along a wellbore axis, the formation having a virgin fluid and a contaminated fluid therein, comprising:

a sample inlet provided on a sample inlet assembly including a sample inlet extension mechanism;

a first guard inlet spaced from the sample inlet in a first direction along the wellbore axis, wherein the first guard inlet is provided on a first guard inlet assembly including a first guard inlet extension mechanism;

a second guard inlet spaced from the sample inlet in a second, opposite direction relative to the first guard inlet along the wellbore axis, wherein the second guard inlet is provided on a second guard inlet assembly including a second guard inlet extension mechanism;

at least one cleanup flowline fluidly connected to the first and second guard inlets for passing the contaminated fluid;

an evaluation flowline fluidly connected to the sample inlet for collecting the virgin fluid;

an inlet packer completely surrounding outer peripheries of the sample inlet, the first guard inlet, and the second guard inlet, wherein the inlet packer includes a first packer segment disposed between the sample inlet and the first guard inlet and a second packer segment disposed between the sample inlet and the second guard inlet;

wherein the sample inlet extension mechanism, the first guard inlet extension mechanism, and the second guard inlet extension mechanism are operable independently of one another.

2. The fluid sampling system of claim **1**, in which the inlet packer has an oval-shaped outer periphery.

3. The fluid sampling system of claim **1**, in which the sample inlet assembly has a diameter, and wherein the first and second guard inlet assemblies are longitudinally spaced from the sample inlet assembly by a distance substantially equal to or greater than the diameter.

4. The fluid sampling system of claim **1**, in which at least one of the first and the second guard inlet assemblies has a diameter, and wherein the at least one of the first and the second guard inlet assembly is longitudinally spaced from the sample inlet assembly by a distance substantially equal to or greater than the diameter.

5. The fluid sampling system of claim **1**, in which the sample inlet assembly, the first guard inlet assembly, and the second guard inlet assembly are provided on a stabilizing blade of a drilling tool.

6. The fluid sampling system of claim **1**, in which the sample inlet is azimuthally offset from the first and the second guard inlets.

7. The fluid sampling system of claim **1**, further comprising a third guard inlet fluidly connected to the cleanup flowline.

8. The fluid sampling system of claim **1**, in which the sample inlet, the first guard inlet, and the second guard inlet are integrally provided an a single probe assembly including an inlet extension mechanism.

9. The fluid sampling system of claim **1**, in which the first and second packer segments further comprise a reinforcement material.

10. The fluid sampling system of claim **1**, in which the sample inlet has an oval-shaped cross-sectional profile, with a major axis perpendicular to the wellbore axis and a minor axis parallel to the wellbore axis.

11. The fluid sampling system of claim **1**, in which the system is associated with a wireline tool.

12. The fluid sampling system of claim **1**, in which the system is associated with a drilling tool.

13. A fluid sampling system for retrieving a formation fluid sample from a formation surrounding a wellbore extending along a wellbore axis, the formation having a virgin fluid and a contaminated fluid therein, comprising:

a sample inlet provided on a sample inlet assembly including a sample inlet extension mechanism;

a first guard inlet spaced from the sample inlet in a first direction along the wellbore axis, wherein the first guard inlet is provided on a first guard inlet assembly including a first guard inlet extension mechanism;

a second guard inlet spaced from the sample inlet in a second, opposite direction relative to the first guard inlet along the wellbore axis, wherein the second guard inlet is provided on a second guard inlet assembly including a second guard inlet extension mechanism;

at least one cleanup flowline fluidly connected to the first and second guard inlets for passing the contaminated fluid; and

an evaluation flowline fluidly connected to the sample inlet for collecting the virgin fluid;

a sample inlet packer completely surrounding an outer periphery of the sample inlet;

a first guard inlet packer completely surrounding an outer periphery of the first guard inlet; and

a second guard inlet packer completely surrounding an outer periphery of the second guard inlet;

wherein the sample inlet extension mechanism, the first guard inlet extension mechanism, and the second guard inlet extension mechanism are operable independently of one another.

14. A fluid sampling system for retrieving a formation fluid sample from a formation surrounding a wellbore extending along a wellbore axis, the formation having a virgin fluid and a contaminated fluid therein, comprising:

a sample inlet provided on a sample inlet assembly including a sample inlet extension mechanism;

a first guard inlet spaced from the sample inlet in a first direction along the wellbore axis, wherein the first guard inlet is provided on a first guard inlet assembly including a first guard inlet extension mechanism;

a second guard inlet spaced from the sample inlet in a second, opposite direction relative to the first guard inlet along the wellbore axis, wherein the second guard inlet

is provided on a second guard inlet assembly including a second guard inlet extension mechanism;

at least one cleanup flowline fluidly connected to the first and the second guard inlets for passing the contaminated fluid;

an evaluation flowline fluidly connected to the sample inlet for collecting the virgin fluid; and

an inlet packer completely surrounding outer peripheries of the sample inlet, the first guard inlet, and the second guard inlet;

in which an exterior face of the inlet packer includes a guard channel; and

wherein the sample inlet extension mechanism, the first guard inlet extension mechanism, and the second guard inlet extension mechanism are operable independently of one another.

15. A probe assembly for use with a fluid sampling system to retrieve a formation fluid sample from a formation surrounding a wellbore extending along a wellbore axis, the formation having a virgin fluid and a contaminated fluid therein, comprising:

an inlet extension mechanism;

a sample inlet coupled to the inlet extension mechanism;

a first guard inlet coupled to the inlet extension mechanism, the first guard inlet being positioned adjacent to the sample inlet and spaced from the sample inlet in a first direction parallel to the wellbore axis;

a second guard inlet coupled to the inlet extension mechanism, the second guard inlet being positioned adjacent to the sample inlet and spaced from the sample inlet in a second, opposite direction relative to the first guard inlet parallel to the wellbore axis; and

an inlet packer completely surrounding outer peripheries of the sample inlet, the first guard inlet, and the second guard inlet;

in which an exterior face of the inlet packer includes a guard channel, and in which the guard channel includes a guard ring section completely surrounding an outer periphery of the first guard inlet and at least a first wing section connected to and extending away from the guard ring section.

16. The probe assembly of claim **15**, in which the guard channel further includes a second wing section connecting to and extending away from the guard ring section.

17. The probe assembly of claim **15**, further including a second guard channel having a second guard ring section completely surrounding an outer periphery of the second guard inlet and at least a second wing section connected to and extending away from the guard ring section.

18. The probe assembly of claim **15**, in which the guard channel is defined by a channel insert coupled to the inlet packer.

19. The probe assembly of claim **18**, in which the channel insert is mechanically coupled to the inlet packer.

20. The probe assembly of claim **15**, in which the sample inlet, the first guard inlet, and the second guard inlet are pivotably coupled to the inlet extension mechanism.

21. A downhole tool connected to a drill string positioned in a wellbore penetrating a subterranean formation along a wellbore axis, the tool comprising:

a drilling collar having at least one stabilizing blade defining a blade axis;

an inlet extension mechanism housed within the stabilizing blade; and

a probe assembly pivotably coupled to the inlet extension mechanism, the probe assembly comprising:

a sample inlet having a mouth portion with a first profile dimension in a direction parallel to the blade axis and a second profile dimension in a direction perpendicular to the blade axis, in which the first profile dimension is greater than the second profile dimension;

an inner packer completely surrounding an outer periphery of the sample inlet;

a guard inlet extending completely around an outer periphery of the inner packer; and

an outer packer completely surrounding an outer periphery of the guard inlet.

22. The downhole tool of claim **21**, in which the mouth portion has a generally oval shape cross-sectional profile, with the first profile dimension comprising a major axis and the second profile dimension comprising a minor axis.

23. The downhole tool of claim **21**, in which the second profile dimension is less than approximately 3.5 inches.

24. A probe assembly for use with a fluid sampling system to retrieve a formation fluid sample from a formation surrounding a wellbore extending along a wellbore axis, the formation having a virgin fluid and a contaminated fluid therein, comprising:

an inlet extension mechanism;

a sample inlet coupled to the inlet extension mechanism;

a first guard inlet coupled to the inlet extension mechanism, the first guard inlet being positioned adjacent to the sample inlet and spaced from the sample inlet in a first direction parallel to the wellbore axis;

a second guard inlet coupled to the inlet extension mechanism, the second guard inlet being positioned adjacent to the sample inlet and spaced from the sample inlet in a second, opposite direction relative to the first guard inlet parallel to the wellbore axis; and

an inlet packer completely surrounding outer peripheries of the sample inlet, the first guard inlet, and the second guard inlet;

wherein an exterior face of the inlet packer includes a guard channel comprising:

a central ring section completely surrounding the outer periphery of the sample inlet;

a first guard ring section completely surrounding the outer periphery of the first guard inlet;

a second guard ring section completely surrounding an outer periphery of the second guard inlet;

a first link section extending between the central ring section and the first guard ring section; and

a second link section extending between the central ring section and the second guard ring section.

25. The probe assembly of claim **24**, in which the guard channel is defined by a channel insert coupled to the inlet packer.

26. The probe assembly of claim **25**, in which the channel insert is mechanically coupled to the inlet packer.

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